

$$1. P(X=0) = \frac{1}{4}, P(X=1) = \frac{1}{2}, P(X=2) = \frac{1}{4}$$

$$P(Y=-1) = \frac{1}{4}, P(Y=0) = \frac{1}{2}, P(Y=1) = \frac{1}{4} \quad \text{不独立}$$

$$2. p(x,y) = \begin{cases} \frac{1}{(b-a)(d-c)}, & \text{当 } (x,y) \in D \text{ 时} \\ 0, & \text{其它} \end{cases} \quad P_X(x) = \int_c^d p(x,y) dy = \frac{1}{b-a}, P_Y(y) = \int_a^b p(x,y) dx = \frac{1}{d-c} \quad \text{独立}$$

$$3. (1) I = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p(x,y) dx dy = \iint_{x^2+y^2 \leq R^2} c(R - \sqrt{x^2+y^2}) dx dy \xrightarrow{\substack{x=r\cos\theta \\ y=r\sin\theta}} \int_0^R dr \int_0^{2\pi} c(R-r)r d\theta$$

$$= 2\pi c \int_0^R (Rr - r^2) dr = 2\pi c \left(\frac{R}{2}r^2 - \frac{1}{3}r^3 \right) \Big|_{r=0}^R = \frac{\pi R^3}{3} c \Rightarrow c = \frac{3}{\pi R^3}$$

$$(2) P(X^2+Y^2 \leq r^2) = \iint_{x^2+y^2 \leq r^2} p(x,y) dx dy \xrightarrow{\substack{x=\rho\cos\theta \\ y=\rho\sin\theta}} \int_0^r d\rho \int_0^{2\pi} c(R-\rho)\rho d\theta$$

$$= 2\pi c \int_0^r (R\rho - \rho^2) d\rho = 2\pi c \left(\frac{R}{2}\rho^2 - \frac{1}{3}\rho^3 \right) \Big|_{\rho=0}^r = 2\pi c r^2 \left(\frac{R}{2} - \frac{r}{3} \right) = \frac{r^2(3R-2r)}{R^3}$$

$$4. \text{该区域是长、短轴在 } y=x \text{ 和 } y=-x \text{ 上, 长度为 } 2a, 2b \text{ 的椭圆. } p(x,y) = \begin{cases} \frac{1}{\pi ab}, & \text{当 } (x,y) \in D \text{ 时} \\ 0, & \text{其它} \end{cases}$$

$$5. (1) I = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p(x,y) dx dy = c \left(\int_{-\infty}^{+\infty} \frac{dx}{1+x^2} \right) \left(\int_{-\infty}^{+\infty} \frac{dy}{1+y^2} \right) = \pi^2 c \Rightarrow c = \frac{1}{\pi^2}$$

$$(2) P_X(x) = \int_{-\infty}^{+\infty} p(x,y) dy = \frac{1}{\pi(1+x^2)}, P_Y(y) = \int_{-\infty}^{+\infty} p(x,y) dx = \frac{1}{\pi(1+y^2)} \quad \text{独立}$$

$$P(0 \leq x \leq 1 \text{ 且 } 0 \leq y \leq 1) = \left(\int_0^1 P_X(x) dx \right) \left(\int_0^1 P_Y(y) dy \right) = \frac{1}{16}$$

$$6. p(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_1}{\sigma_1} \right)^2 - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \left(\frac{y-\mu_2}{\sigma_2} \right)^2 \right] \right\}$$

$$\Rightarrow P_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp \left\{ -\frac{(x-\mu_1)^2}{2\sigma_1^2} \right\}, P_Y(y) = \frac{1}{\sqrt{2\pi}\sigma_2} \exp \left\{ -\frac{(y-\mu_2)^2}{2\sigma_2^2} \right\}, \rho_{XY} = \rho$$

$$(1) p(x,y) = \frac{1}{\sqrt{3}\pi} \exp \left\{ -\frac{2}{3} [(x-3)^2 - (x-3)y + y^2] \right\}$$

$$(2) p(x,y) = \frac{4}{\sqrt{3}\pi} \exp \left\{ -\frac{2}{3} [4(x-1)^2 + 4(x-1)(y-1) + 4(y-1)^2] \right\}$$

$$(3) p(x,y) = \frac{1}{\pi} \exp \left\{ -\frac{1}{2} [(x-1)^2 + 4(y-2)^2] \right\}$$

$$7. \text{设 } X+Y=Z, \text{ 则 } p_2(z) = \int_{-\infty}^{+\infty} p_X(z-u)p_Y(u) du. \text{ 当 } z \leq 0 \text{ 时 } p_2(z) = 0$$

$$\text{当 } 0 < z \leq 1 \text{ 时 } p_2(z) = \int_0^z p_X(z-u)p_Y(u) du = \int_0^z e^{-u} du = 1 - e^{-z}$$

$$\text{当 } z > 1 \text{ 时 } p_2(z) = \int_{z-1}^z p_X(z-u)p_Y(u) du = \int_{z-1}^z e^{-u} du = e^{-(z-1)} - e^{-z}$$

$$8. \text{设 } X+Y=Z, \text{ 则 } p_2(z) = \int_{-\infty}^{+\infty} p_X(z-u)p_Y(u) du. \text{ 当 } z \leq 0 \text{ 时 } p_2(z) = 0$$

$$\text{当 } z > 0 \text{ 时 } p_2(z) = \int_0^z p_X(z-u)p_Y(u) du = \int_0^z \frac{1}{2^{\frac{m}{2}}\Gamma(\frac{m}{2})} (z-u)^{\frac{m}{2}-1} e^{-\frac{z-u}{2}} \cdot \frac{1}{2^{\frac{n}{2}}\Gamma(\frac{n}{2})} u^{\frac{n}{2}-1} e^{-\frac{u}{2}} du$$

$$= \frac{e^{-\frac{z}{2}}}{2^{\frac{m+n}{2}} \Gamma(\frac{m}{2}) \Gamma(\frac{n}{2})} \int_0^z u^{\frac{n}{2}-1} (z-u)^{\frac{m}{2}-1} du \stackrel{u=zt}{=} \frac{e^{-\frac{z}{2}}}{2^{\frac{m+n}{2}} \Gamma(\frac{m}{2}) \Gamma(\frac{n}{2})} \int_0^1 z^{\frac{n}{2}+\frac{m}{2}-2} t^{\frac{n}{2}-1} (1-t)^{\frac{m}{2}-1} z dt$$

$$= \frac{e^{-\frac{z}{2}}}{2^{\frac{m+n}{2}} \Gamma(\frac{m}{2}) \Gamma(\frac{n}{2})} \cdot z^{\frac{n}{2}+\frac{m}{2}-1} \underbrace{\int_0^1 t^{\frac{n}{2}-1} (1-t)^{\frac{m}{2}-1} dt}_{\text{B函数的定义, 它就是 } B(\frac{n}{2}, \frac{m}{2})}$$

$$= \frac{e^{-\frac{z}{2}}}{2^{\frac{m+n}{2}} \Gamma(\frac{m}{2}) \Gamma(\frac{n}{2})} \cdot z^{\frac{n}{2}+\frac{m}{2}-1} \cdot \frac{\Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})}{\Gamma(\frac{n}{2}+\frac{m}{2})} = \frac{1}{2^{\frac{m+n}{2}} \Gamma(\frac{m+n}{2})} z^{\frac{m+n}{2}-1} e^{-\frac{z}{2}}$$

9. 1) $Z = \min\{X, Y\}$. 当 $z \leq 0$ 时 $p_z(z) = 0$

当 $z > 0$ 时 $P(Z \leq z) = P(X \leq z \text{ 或 } Y \leq z) = 1 - P(X > z \text{ 且 } Y > z)$

$$= 1 - (\int_z^{+\infty} \alpha e^{-\alpha x} dx) (\int_z^{+\infty} \beta e^{-\beta y} dy) = 1 - e^{-(\alpha+\beta)z} \text{ 求导得 } p_z(z) = (\alpha+\beta)e^{-(\alpha+\beta)z}$$

(2) $Z = \max\{X, Y\}$. 当 $z \leq 0$ 时 $p_z(z) = 0$

当 $z > 0$ 时 $P(Z \leq z) = P(X \leq z \text{ 且 } Y \leq z) = (\int_0^z \alpha e^{-\alpha x} dx) (\int_0^z \beta e^{-\beta y} dy)$

$$= (1 - e^{-\alpha z})(1 - e^{-\beta z}) \text{ 求导得 } p_z(z) = \alpha e^{-\alpha z} + \beta e^{-\beta z} - (\alpha+\beta)e^{-(\alpha+\beta)z}$$

(3) $Z = X + Y$. 当 $z \leq 0$ 时 $p_z(z) = 0$

$$\text{当 } z > 0 \text{ 时 } p_z(z) = \int_{-\infty}^{+\infty} p_x(u) p_y(z-u) du = \int_0^z \alpha e^{-\alpha u} \cdot \beta e^{-\beta(z-u)} du$$

$$= \alpha \beta e^{-\beta z} \int_0^z e^{(\beta-\alpha)u} du = \frac{\alpha \beta}{\beta-\alpha} e^{-\beta z} (e^{(\beta-\alpha)z} - 1) = \frac{\alpha \beta}{\beta-\alpha} (e^{-\alpha z} - e^{-\beta z})$$

10. 当 $z \leq 0$ 时 $p_z(z) = 0$. 当 $z > 0$ 时 $P(Z \leq z) = P(X^2 + Y^2 \leq z^2) = \iint_{\substack{x^2+y^2 \leq z^2 \\ x>0, y>0}} p(x, y) dx dy$

$$\stackrel{\substack{x=r\cos\theta \\ y=r\sin\theta}}{=} \int_0^z dr \int_0^{\frac{\pi}{2}} 4(r\cos\theta)(r\sin\theta) e^{-r^2} r d\theta = (\int_0^z r^3 e^{-r^2} dr) (\int_0^{\frac{\pi}{2}} 2\sin 2\theta d\theta)$$

$$= 2 \int_0^z r^3 e^{-r^2} dr = \int_0^z (-r^2) d(e^{-r^2}) = (-r^2 e^{-r^2}) \Big|_{r=0}^z + \int_0^z 2r e^{-r^2} dr = 1 - (z^2+1)e^{-z^2}$$

$$\text{求导得 } p_z(z) = -(z^2+1) \cdot (-2z) e^{-z^2} - 2z e^{-z^2} = 2z^3 e^{-z^2} \quad (z > 0)$$

$$\text{则 } E(Z) = \int_0^{+\infty} z \cdot 2z^3 e^{-z^2} dz \stackrel{z=t^{\frac{1}{2}}}{=} \int_0^{+\infty} 2t^2 e^{-t} \cdot \frac{1}{2\sqrt{t}} dt = \Gamma(\frac{5}{2}) = \frac{3\sqrt{\pi}}{4}$$

$$11. p_x(x) = \int_0^x p(x, y) dy = 2x \quad E(X) = \int_0^1 x \cdot 2x dx = \frac{2}{3}$$

$$E(X^2) = \int_0^1 x^2 \cdot 2x dx = \frac{1}{2} \quad \text{var}(X) = E(X^2) - (E(X))^2 = \frac{1}{18}$$

$$p_y(y) = \int_y^1 p(x, y) dx = 2(1-y) \quad E(Y) = \int_0^1 y \cdot 2(1-y) dy = \frac{1}{3}$$

$$E(Y^2) = \int_0^1 y^2 \cdot 2(1-y) dy = \frac{1}{6} \quad \text{var}(Y) = E(Y^2) - (E(Y))^2 = \frac{1}{18}$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy p(x,y) dx dy = \int_0^1 x dx \int_0^1 2y dy = \frac{1}{4}$$

$$E((X-E(X))(Y-E(Y))) = E(XY) - E(X)E(Y) = \frac{1}{36} \quad \rho_{XY} = \frac{E((X-E(X))(Y-E(Y)))}{\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)}} = \frac{1}{2}$$

12. n 为偶数时. $\text{cov}(X, Y) = \underbrace{E(X^{n+1})}_{=0} - \underbrace{E(X)}_{=0} E(X^n) = 0$. 则 $\rho_{XY} = 0$

n 为奇数时. $\text{cov}(X, Y) = E(X^{n+1}) - E(X)E(X^n) = E(X^{n+1})$

而 m 为偶数时, $E(X^m) = \int_{-\infty}^{+\infty} x^m \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \sqrt{\frac{2}{\pi}} \int_0^{+\infty} x^m e^{-\frac{x^2}{2}} dx$

$$\stackrel{x=\sqrt{2}t}{=} \sqrt{\frac{2}{\pi}} \int_0^{+\infty} 2^{\frac{m}{2}} t^{\frac{m}{2}} e^{-t} \cdot \sqrt{2} \cdot \frac{1}{2\sqrt{t}} dt = \frac{2^{\frac{m}{2}}}{\sqrt{\pi}} \int_0^{+\infty} t^{\frac{m-1}{2}} e^{-t} dt = \frac{2^{\frac{m}{2}}}{\sqrt{\pi}} \Gamma(\frac{m+1}{2})$$

$$= \frac{2^{\frac{m}{2}}}{\sqrt{\pi}} \cdot \frac{m-1}{2} \cdot \frac{m-3}{2} \cdots \frac{1}{2} \cdot \sqrt{\pi} = \frac{(m)!}{(m)!!} = \frac{(m)!}{2^{\frac{m}{2}} (\frac{m}{2})!}$$

因此 $E(X^{2n}) = \frac{(2n)!}{2^n n!}$. $\text{var}(X^n) = E(X^{2n}) - (E(X^n))^2 = \frac{(2n)!}{2^n n!}$

$$\rho_{XY} = \frac{E(X^{n+1})}{\sqrt{\text{var}(X)}\sqrt{\text{var}(X^n)}} = \frac{(n+1)!}{2^{\frac{n+1}{2}} (\frac{n+1}{2})!} \cdot \frac{2^{\frac{n}{2}} \sqrt{n!}}{\sqrt{(2n)!}} = \frac{n!!}{\sqrt{(2n-1)!!}}$$

13. $E(X_1 X_2) = \int_0^1 dx_1 \int_5^{+\infty} dx_2 \cdot 2x_1 e^{-(x_2-5)} \cdot x_1 x_2 = (\int_0^1 2x_1^2 dx_1) (\int_5^{+\infty} x_2 e^{-(x_2-5)} dx_2) = 4$

14. $E((X-E(X))(Y-E(Y))) = \rho_{XY} \sqrt{\text{var}(X)} \sqrt{\text{var}(Y)} = 12$

$$\text{var}(X+Y) = E((X-E(X)+Y-E(Y))^2) = E((X-E(X))^2) + 2E((X-E(X))(Y-E(Y))) + E((Y-E(Y))^2) = 85$$

$$\text{var}(X-Y) = E((X-E(X)-Y+E(Y))^2) = E((X-E(X))^2) - 2E((X-E(X))(Y-E(Y))) + E((Y-E(Y))^2) = 37$$

15. $p(x,y) = \frac{1}{2\pi ab} e^{-\frac{1}{2}(\frac{x^2}{a^2} + \frac{y^2}{b^2})}$. $P((X,Y) \in D) = \iint_{(x,y) \in D} \frac{1}{2\pi ab} e^{-\frac{1}{2}(\frac{x^2}{a^2} + \frac{y^2}{b^2})} dx dy$

$$\stackrel{\substack{x=r\cos\theta \\ y=r\sin\theta}}{=} \int_0^k dr \int_0^{2\pi} d\theta \cdot \frac{1}{2\pi ab} e^{-\frac{1}{2}r^2} \cdot abr = \int_0^k r e^{-\frac{1}{2}r^2} dr = 1 - e^{-\frac{k^2}{2}}$$

16. $P_X(x) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p(x,y,z) dy dz = e^{-x}$ 其它两个同理. 相互独立

17. $t < 0$ 时 $P_3(t) = 0$. $t \geq 0$ 时 $P(Z \leq t) = \iiint_{x^2+y^2+z^2 \leq t^2} p_X(x)p_Y(y)p_Z(z) dx dy dz$

$$= \iiint_{x^2+y^2+z^2 \leq t^2} \frac{1}{(2\pi)^{\frac{3}{2}}} e^{-\frac{x^2+y^2+z^2}{2}} dx dy dz \stackrel{\substack{x=r\cos\theta\cos\varphi \\ y=r\cos\theta\sin\varphi \\ z=r\sin\theta}}{=} \int_0^t dr \int_0^\pi d\theta \int_0^{2\pi} d\varphi \frac{1}{(2\pi)^{\frac{3}{2}}} e^{-\frac{r^2}{2}} \cdot r^2 \sin\theta$$

$$= \frac{1}{\sqrt{2\pi}} \left(\int_0^\pi \sin\theta d\theta \right) \left(\int_0^t r^2 e^{-\frac{r^2}{2}} dr \right) = \sqrt{\frac{2}{\pi}} \left(-te^{-\frac{t^2}{2}} + \int_0^t e^{-\frac{r^2}{2}} dr \right) \quad \text{求导得 } P_3(t) = \sqrt{\frac{2}{\pi}} t^2 e^{-\frac{t^2}{2}}$$

$$18. t > 0 \text{ 时 } P(\xi \leq t) = P(\min\{X_1, \dots, X_n\} \leq t) = 1 - P(X_1 > t) \cdots P(X_n > t) \\ = 1 - e^{-\left(\frac{t}{\eta}\right)^m} \cdots e^{-\left(\frac{t}{\eta}\right)^m} = 1 - e^{-n\left(\frac{t}{\eta}\right)^m} = 1 - e^{-\left(\frac{t}{\eta \cdot \left(\frac{1}{n}\right)^{\frac{1}{m}}}\right)^m}$$

$$19. E(X+Y+Z) = E(X) + E(Y) + E(Z) = 1, \text{COV}(Y, Z) = \rho_{YZ} \sqrt{\text{var}(Y)} \sqrt{\text{var}(Z)} = -\frac{1}{2}$$

$$\text{COV}(X, Y) = \rho_{XY} \sqrt{\text{var}(X)} \sqrt{\text{var}(Y)} = 0, \text{COV}(X, Z) = \rho_{XZ} \sqrt{\text{var}(X)} \sqrt{\text{var}(Z)} = \frac{1}{2}$$

$$\text{var}(X+Y+Z) = E\left((X-E(X)+Y-E(Y)+Z-E(Z))^2\right)$$

$$= \text{var}(X) + \text{var}(Y) + \text{var}(Z) + 2\text{COV}(X, Y) + 2\text{COV}(Y, Z) + 2\text{COV}(X, Z) = 3$$

$$20. p_{u,v}(u, v) = p_{x,y}(x, y) \left| \frac{D(x, y)}{D(u, v)} \right| = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{u+v}{2}\right)^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{u-v}{2}\right)^2} \cdot \frac{1}{2} = \frac{1}{4\pi} e^{-\frac{1}{4}(u^2+v^2)}$$

$$21. \text{注意这题 } U = X^2 + Y^2, V = \frac{X}{Y} \Rightarrow \begin{cases} X = V \sqrt{\frac{U}{1+V^2}}, Y = \sqrt{\frac{U}{1+V^2}} \\ X = -V \sqrt{\frac{U}{1+V^2}}, Y = -\sqrt{\frac{U}{1+V^2}} \end{cases} \text{可以解出两组}(X, Y)$$

我们想用 $p_{u,v}(u, v) = p_{x,y}(x, y) \left| \frac{D(x, y)}{D(u, v)} \right|$, 但这要求 (X, Y) 与 (u, v) 是一一对应的, 所以不能用

想一个曲线救国的方法: 极坐标变换. 令 $X = R \cos \theta, Y = R \sin \theta$

这时候 (X, Y) 与 (R, θ) 是一一对应的, 所以可以用上面那个公式

$$p_{R,\theta}(R, \theta) = p_{x,y}(x, y) \left| \frac{D(x, y)}{D(R, \theta)} \right| = \frac{R}{2\pi} e^{-\frac{R^2}{2}}$$

$$\text{而 } p_R(R) = \int_0^{2\pi} p_{R,\theta}(R, \theta) d\theta = R e^{-\frac{R^2}{2}}, p_\theta(\theta) = \int_0^{+\infty} p_{R,\theta}(R, \theta) dR = \frac{1}{2\pi}$$

所以 $p_{R,\theta}(R, \theta) = p_R(R) \cdot p_\theta(\theta)$, 即 R 与 θ 独立

而 $U = X^2 + Y^2 = R^2, V = \frac{X}{Y} = \cot \theta$, 所以 U 与 V 独立

$$22. E(XY) = \int_0^1 dx \int_1^3 dy \cdot xy \cdot 1 \cdot \frac{1}{2} = 1$$

$$E(X^2 Y^2) = \int_0^1 dx \int_1^3 dy \cdot x^2 y^2 \cdot 1 \cdot \frac{1}{2} = \frac{13}{9}, \text{var}(XY) = E(X^2 Y^2) - (E(XY))^2 = \frac{4}{9}$$

$$23. \text{var}(XY) - \text{var}(X) \text{var}(Y) = E(X^2 Y^2) - (E(XY))^2 - (E(X^2) - (E(X))^2)(E(Y^2) - (E(Y))^2)$$

$$= E(X^2)E(Y^2) - (E(X)E(Y))^2 - (E(X^2) - (E(X))^2)(E(Y^2) - (E(Y))^2)$$

$$= (E(X))^2(E(Y^2) - (E(Y))^2) + (E(Y))^2(E(X^2) - (E(X))^2) \geq 0$$

$$24. E(X_1) = \cdots = E(X_r) = \sum_j \frac{n_j x_j}{n} = m \Rightarrow E(X_1 + \cdots + X_r) = mr$$

$$\text{var}(X_1 + \dots + X_r) = E\left((X_1 - E(X_1) + \dots + X_r - E(X_r))^2\right) = \sum_{i=1}^r E((X_i - E(X_i))^2) + 2 \sum_{1 \leq i < j \leq r} E((X_i - E(X_i))(X_j - E(X_j)))$$

$$E(X_i^2) = \sum_j \frac{n_j x_j^2}{n} = m^2 + \sigma^2$$

$$E((X_i - E(X_i))^2) = E((X_i - E(X_i))^2) = E(X_i^2) - (E(X_i))^2 = \sigma^2$$

$E(X_1 X_2)$. 它的定义就是 $X_1 X_2$ 所有取值乘以概率再求和

设这 n 个区居民数为 $y_1, \dots, y_n$ (就是 $n_1 \uparrow x_1, n_2 \uparrow x_2, \dots, n_t \uparrow x_t$)

$$\text{那么 } E(X_1 X_2) = \frac{1}{C_n^2} \sum_{1 \leq i < j \leq n} y_i y_j = \frac{1}{n(n-1)} \left[\left(\sum_{i=1}^n y_i \right)^2 - \sum_{i=1}^n y_i^2 \right]$$

$$= \frac{1}{n(n-1)} \left[\left(\sum_j n_j x_j \right)^2 - \sum_j n_j x_j^2 \right] = \frac{1}{n(n-1)} \left[n^2 m^2 - n(\sigma^2 + m^2) \right]$$

$$E((X_i - E(X_i))(X_j - E(X_j))) = E((X_i - E(X_i))(X_j - E(X_j))) = E(X_i X_j) - (E(X_i))(E(X_j)) = -\frac{\sigma^2}{n-1}$$

$$\text{代回原式得 } \text{var}(X_1 + \dots + X_r) = r\sigma^2 + 2 \cdot \frac{1}{2} r(r-1) \left(-\frac{\sigma^2}{n-1} \right) = \frac{\sigma^2 r(n-r)}{n-1}$$

$$25. \quad \forall 1 \leq i_1 < \dots < i_k \leq n, \text{ 有 } E\left(\frac{X_{i_1} + \dots + X_{i_k}}{X_1 + \dots + X_n}\right) = E\left(\frac{X_1 + \dots + X_k}{X_1 + \dots + X_n}\right)$$

$$\text{因此 } E\left(\frac{X_1 + \dots + X_k}{X_1 + \dots + X_n}\right) = \frac{1}{C_n^k} \sum_{1 \leq i_1 < \dots < i_k \leq n} E\left(\frac{X_{i_1} + \dots + X_{i_k}}{X_1 + \dots + X_n}\right) = \frac{C_{n-1}^{k-1}}{C_n^k} = \frac{k}{n}$$

$$26. \text{ 首先有个结论. 若 } X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2). \text{ 则 } X+Y \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$$

$$\text{所以 } \xi \sim N(m\mu, m\sigma^2), \eta \sim N(n\mu, n\sigma^2)$$

然后有个结论, 若 $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$, 则 (X, Y) 的联合分布是二元正态分布.

而二元正态分布有 $(\mu_1, \mu_2, \sigma_1, \sigma_2, \rho)$ 五个量需要确定. 给出 X, Y 后前4个直接

可以得出来. 而 ρ 是由 $\rho = \rho_{XY}$ 确定的. 这样我们就给出了 (X, Y) 的联合分布

我们写得专业一点, 把多元正态分布用矩阵来写:

$$\vec{x} = \begin{bmatrix} \xi \\ \eta \end{bmatrix} \text{ 的联合密度为 } p(\vec{x}) = \frac{1}{(2\pi)^{\frac{1}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} (\vec{x} - \vec{\mu})^T \Sigma^{-1} (\vec{x} - \vec{\mu}) \right\}$$

$$\text{其中 } \vec{\mu} \text{ 为均值 } \begin{bmatrix} m\mu \\ n\mu \end{bmatrix}. \Sigma \text{ 为协方差矩阵 } \begin{bmatrix} \text{cov}(\xi, \xi) & \text{cov}(\xi, \eta) \\ \text{cov}(\eta, \xi) & \text{cov}(\eta, \eta) \end{bmatrix} = \begin{bmatrix} m\sigma^2 & m\sigma^2 \\ m\sigma^2 & n\sigma^2 \end{bmatrix}$$

$$(\text{cov}(\xi, \eta) = E((X_1 - E(X_1) + \dots + X_m - E(X_m))(X_1 - E(X_1) + \dots + X_n - E(X_n))) = m\sigma^2)$$

$$\text{计算它的逆 } \Sigma^{-1} = \frac{1}{\sigma^2} \cdot \frac{1}{mn-m^2} \begin{bmatrix} n & -m \\ -m & m \end{bmatrix} = \begin{bmatrix} \frac{n}{\sigma^2 m(n-m)} & -\frac{1}{\sigma^2(n-m)} \\ -\frac{1}{\sigma^2(n-m)} & \frac{1}{\sigma^2(n-m)} \end{bmatrix}$$

$$(\vec{x}-\vec{\mu})^T \Sigma^{-1} (\vec{x}-\vec{\mu}) = [z-m\mu, \eta-n\mu] \begin{bmatrix} \frac{n}{\sigma^2 m(n-m)} & -\frac{1}{\sigma^2(n-m)} \\ -\frac{1}{\sigma^2(n-m)} & \frac{1}{\sigma^2(n-m)} \end{bmatrix} \begin{bmatrix} z-m\mu \\ \eta-n\mu \end{bmatrix}$$

$$= \frac{n}{\sigma^2 m(n-m)} \cdot (z-m\mu)^2 - \frac{2}{\sigma^2(n-m)} (z-m\mu)(\eta-n\mu) + \frac{1}{\sigma^2(n-m)} \cdot (\eta-n\mu)^2$$

$|\Sigma| = (mn-m^2)\sigma^2$. 再把以上所有东西代回原式就行

$$27. (1) E(\alpha X + \beta Y) = (\alpha + \beta)\mu, E(\alpha X - \beta Y) = (\alpha - \beta)\mu$$

$$X \text{ 与 } Y \text{ 独立} \Rightarrow \alpha X \text{ 与 } \beta Y \text{ 独立} \Rightarrow \text{var}(\alpha X \pm \beta Y) = \text{var}(\alpha X) + \text{var}(\beta Y) = (\alpha^2 + \beta^2)\sigma^2$$

$$E((\alpha X + \beta Y)(\alpha X - \beta Y)) = E(\alpha^2 X^2) - E(\beta^2 Y^2) = (\alpha^2 - \beta^2)(\sigma^2 + \mu^2)$$

$$\text{设 } U = \alpha X + \beta Y, V = \alpha X - \beta Y. \text{ 则 } X = \frac{U+V}{2\alpha}, Y = \frac{U-V}{2\beta}$$

$$P_{U,V}(u,v) = P_{X,Y}(x,y) \left| \frac{\partial D(x,y)}{\partial D(u,v)} \right| = \frac{1}{2\alpha\beta} \cdot \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{1}{2\sigma^2} \left[\left(\frac{u+v}{2\alpha} \right)^2 + \left(\frac{u-v}{2\beta} \right)^2 \right]}$$

$$\rho_{U,V} = \frac{\text{cov}(U,V)}{\sqrt{\text{var}(U)} \cdot \sqrt{\text{var}(V)}} = \frac{(\alpha^2 - \beta^2)\sigma^2}{\sqrt{(\alpha^2 + \beta^2)\sigma^2} \cdot \sqrt{(\alpha^2 + \beta^2)\sigma^2}} = \frac{\alpha^2 - \beta^2}{\alpha^2 + \beta^2}$$

$$(2) \text{ 设 } Z = \max\{X, Y\}. \text{ 则 } P(Z \leq z) = P(X \leq z \text{ 且 } Y \leq z) = \left(\int_{-\infty}^z \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \right)^2$$

$$\text{求导得 } P_Z(z) = 2 \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \left(\int_{-\infty}^z \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \right)$$

$$\text{则 } E(Z) = 2 \int_{-\infty}^{+\infty} z \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \left(\int_{-\infty}^z \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \right)$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\pi\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \left(\int_x^{+\infty} z e^{-\frac{(z-\mu)^2}{2\sigma^2}} dz \right)$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\pi\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \cdot \left[\int_x^{+\infty} (z-\mu+\mu) e^{-\frac{(z-\mu)^2}{2\sigma^2}} dz \right]$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\pi\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \left(\int_x^{+\infty} (z-\mu) e^{-\frac{(z-\mu)^2}{2\sigma^2}} dz \right) + \int_{-\infty}^{+\infty} \frac{1}{\pi\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \left(\int_x^{+\infty} \mu e^{-\frac{(z-\mu)^2}{2\sigma^2}} dz \right)$$

$$\text{而 } \int_x^{+\infty} (z-\mu) e^{-\frac{(z-\mu)^2}{2\sigma^2}} dz = -\sigma^2 e^{-\frac{(z-\mu)^2}{2\sigma^2}} \Big|_{z=x}^{+\infty} = \sigma^2 e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \text{ 所以}$$

$$\text{前面那一项} = \int_{-\infty}^{+\infty} \frac{1}{\pi} e^{-\frac{(x-\mu)^2}{\sigma^2}} dx \stackrel{x=\mu+\sigma y}{=} \int_{-\infty}^{+\infty} \frac{1}{\pi} e^{-y^2} \sigma dy = \frac{\sigma}{\sqrt{\pi}}$$

$$\text{后面那一项} = \iint_{z \geq x} \frac{\mu}{\pi\sigma^2} e^{-\frac{(x-\mu)^2 + (z-\mu)^2}{2\sigma^2}} dx dz \quad \text{由于对称性,把 } x \text{ 和 } z \text{ 互换是相等的}$$

$$= \frac{1}{2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\mu}{\pi \sigma^2} e^{-\frac{(x-\mu)^2 + (z-\mu)^2}{2\sigma^2}} dx dz = \frac{\mu}{2\pi\sigma^2} \left(\int_{-\infty}^{+\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \right)^2 = \mu$$

28. (1) 当 $\lambda_1 + \dots + \lambda_r = m$ 时 $P(X=\lambda_1, \dots, X_r=\lambda_r) = C_m^{\lambda_1} p_1^{\lambda_1} C_{m-\lambda_1}^{\lambda_2} p_2^{\lambda_2} \dots C_{m-\lambda_1-\dots-\lambda_{r-1}}^{\lambda_r} p_r^{\lambda_r}$

$$= \frac{m!}{\lambda_1!(m-\lambda_1)!} \frac{(m-\lambda_1)!}{\lambda_2!(m-\lambda_1-\lambda_2)!} \dots \frac{(m-\dots-\lambda_{r-1})!}{\lambda_r!(m-\dots-\lambda_r)!} p_1^{\lambda_1} \dots p_r^{\lambda_r} = \frac{m!}{\lambda_1! \dots \lambda_r!} p_1^{\lambda_1} \dots p_r^{\lambda_r}$$

当 $\lambda_1 + \dots + \lambda_r \neq m$ 时 $P(X=\lambda_1, \dots, X_r=\lambda_r) = 0$

(2) $\text{cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$. $E(X_i) = mp_i$, $E(X_j) = mp_j$

$$E(X_i X_j) = \sum_{u=0}^m \sum_{v=0}^{m-u} P(X_i=u, X_j=v) uv = \sum_{u=0}^m \sum_{v=0}^{m-u} \frac{m!}{u!v!(m-u-v)!} uv$$

用上面这个方法也可以强行算出来, 但我们再想个好办法, 可以完全避免计算

设随机变量 $A_k = \begin{cases} 1, & \text{若第 } k \text{ 次试验为结果 } i \\ 0, & \text{其它} \end{cases}$, $B_k = \begin{cases} 1, & \text{若第 } k \text{ 次试验为结果 } j \\ 0, & \text{其它} \end{cases}$

则 $X_i = \sum_{k=1}^m A_k$, $X_j = \sum_{k=1}^m B_k$. 当 $u \neq v$ 时 A_u 和 B_v 是独立的, $\text{cov}(A_u, B_v) = 0$

于是 $\text{cov}(X_i, X_j) = \sum_{u=1}^m \sum_{v=1}^m \text{cov}(A_u, B_v) = \sum_{k=1}^m \text{cov}(A_k, B_k) = n(E(A_k B_k) - (E(A_k))(E(B_k))) = -mp_i p_j$

上面最后一个等号是因为 A 与 B 不可能同时为 1, 所以 $E(A_k B_k) = 0$

29. 要证 $\bar{X}$ 与 η 不相关, 其实就是验证 $E(\bar{X}\eta) = E(\bar{X})E(\eta)$

$$E(\bar{X}) = \frac{1}{n} \cdot n E(X_1) = E(X_1)$$

$$E(\eta) = E\left(\sum_{i=1}^n (X_i^2 - 2\bar{X}X_i + \bar{X}^2)\right) = E\left(\sum_{i=1}^n X_i^2 - \frac{1}{n}\left(\sum_{i=1}^n X_i\right)^2\right)$$

$$= E\left(\frac{n-1}{n} \sum_{i=1}^n X_i^2 - \frac{2}{n} \sum_{1 \leq i < j \leq n} X_i X_j\right) = (n-1)E(X_1^2) - (n-1)(E(X_1))^2$$

$$E(\bar{X}\eta) = E\left(\left(\frac{1}{n} \sum_{i=1}^n X_i\right)\left(\sum_{i=1}^n X_i^2 - \frac{1}{n}\left(\sum_{i=1}^n X_i\right)^2\right)\right) = \frac{1}{n^2} E\left(n\left(\sum_{i=1}^n X_i\right)\left(\sum_{i=1}^n X_i^2\right) - \left(\sum_{i=1}^n X_i\right)^3\right)$$

$$= \frac{1}{n^2} E\left(n \sum_{i=1}^n X_i^3 + n \sum_{1 \leq i < j \leq n} (X_i^2 X_j + X_i X_j^2) - \sum_{i=1}^n X_i^3 - 3 \sum_{1 \leq i < j \leq n} (X_i^2 X_j + X_i X_j^2) - 6 \sum_{1 \leq i < j < k \leq n} X_i X_j X_k\right)$$

$$= \frac{1}{n^2} \left(n(n-1)E(X_1^3) + (n-3)n(n-1)E(X_1^2)E(X_1) - n(n-1)(n-2)(E(X_1))^3 \right)$$

再验证 $E(\bar{X}\eta) - E(\bar{X})E(\eta) = 0$ 即可.

代数变形的主旨是要利用 X_i 独立同分布, 也就是 $E(X_i X_j) = E(X_i)E(X_j)$.

必须要把表达式拆成纯粹的交叉项

注意 X 与 Y 独立是能推出 $f(X)$ 与 $g(Y)$ 独立的 (所以 $E(X_i^2 X_j) = E(X_i^2)E(X_j)$)

变形的时候时刻检查项数, 这样就不会算错

30. $\text{cov}(|X|, X) = E(|X|X) - E(|X|)E(X)$. 由 X 的分布函数是偶函数

知 $E(X) = 0$, $E(|X|X) = 0$. 因此 $\text{cov}(|X|, X) = 0$

不独立是因为当 $|u| \neq |v|$ 时 $P(|X|=u \text{ 且 } X=v) = 0$, 但 $P(|X|=u)P(X=v)$ 可能不为 0

$$\text{var}\left(\sum_{i=1}^n a_i X_i\right) = E\left(\left(\sum_{i=1}^n a_i (X_i - E(X_i))\right)^2\right) = \sum_{i=1}^n E\left(a_i^2 (X_i - E(X_i))^2\right) = \sum_{i=1}^n a_i^2 \sigma_i^2 \geq \frac{\left(\sum_{i=1}^n a_i\right)^2}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}$$

由柯西的取等条件: $\frac{a_i^2 \sigma_i^2}{\frac{1}{\sigma_i^2}} = \dots = \frac{a_n^2 \sigma_n^2}{\frac{1}{\sigma_n^2}} \Rightarrow a_i = \frac{\frac{1}{\sigma_i^2}}{\frac{1}{\sigma_1^2} + \dots + \frac{1}{\sigma_n^2}} \quad (1 \leq i \leq n)$

31.

$\begin{matrix} & X \\ Y & \end{matrix}$	x_1	x_2
y_1	p_{11}	p_{21}
y_2	p_{12}	p_{22}

$$\text{不相关} \Rightarrow (p_{11} + p_{12})x_1 + (p_{21} + p_{22})x_2 \left((p_{11} + p_{21})y_1 + (p_{12} + p_{22})y_2 \right) = p_{11}x_1y_1 + p_{12}x_1y_2 + p_{21}x_2y_1 + p_{22}x_2y_2$$

$$\text{记 } P(X=x_1)=p_1, P(X=x_2)=p_2, P(Y=y_1)=q_1, P(Y=y_2)=q_2$$

$$\text{则不相关} \Rightarrow (p_1x_1 + p_2x_2)(q_1y_1 + q_2y_2) = p_{11}x_1y_1 + p_{12}x_1y_2 + p_{21}x_2y_1 + p_{22}x_2y_2$$

分析一下, 我们想证明 $p_{11}=p_1q_1, p_{12}=p_1q_2, p_{21}=p_2q_1, p_{22}=p_2q_2$

所以就是, $x_1, x_2, y_1, y_2, p_1, p_2, q_1, q_2$ 当已知量, 去解 $p_{11}, p_{12}, p_{21}, p_{22}$

除了上面那个很长的等式, 我们还有 $p_{11} + p_{12} + p_{21} + p_{22} = 1, p_1 = p_{11} + p_{12}, q_1 = p_{11} + p_{21}$

四个方程四个未知数, 刚好够, 所以代入消元一定是可做的.

有了这个信心, 我们来实操一下.

$$(p_1x_1 + (1-p_1)x_2)(q_1y_1 + (1-q_1)y_2) = p_{11}x_1y_1 + (p_1 - p_{11})x_1y_2 + (q_1 - p_{11})x_2y_1 + (1 - p_1 - q_1 + p_{11})x_2y_2$$

$$(p_1q_1 - p_{11})x_1y_1 + (p_1 - p_1q_1)x_1y_2 + (p_{11} - p_1q_1)x_2y_1 + (p_1q_1 - p_{11})x_2y_2 = 0$$

$$(p_1q_1 - p_{11})(x_1 - x_2)(y_1 - y_2) = 0 \Rightarrow p_1q_1 = p_{11}. \text{ 其它三个同理.}$$

所以说, 在做题的时候思路一定要清晰, 时刻明确目标是什么.

以目标为导向, 我们就能很自信地往下做更进一步的尝试

$$32. (1) P(X+Y=0)=0.3. P(X+Y=1)=0.6. P(X+Y=2)=0.1$$

$$(2) P(X-Y=0)=0.4. P(X-Y=-1)=0.3. P(X-Y=1)=0.3$$

$$(3) P(Z=4)=0.3. P(Z=6)=0.3. P(Z=7)=0.3. P(Z=9)=0.1$$

$$E(Z)=4 \cdot 0.3 + 6 \cdot 0.3 + 7 \cdot 0.3 + 9 \cdot 0.1 = 6$$

$$33. \text{ 若 } E(\delta(a-X)\delta(b-Y)) = E(\delta(a-X))E(\delta(b-Y))$$

$$\text{注意到 } E(\delta(a-X)\delta(b-Y)) = P(X \leq a \text{ 且 } Y \leq b), E(\delta(a-X)) = P(X \leq a), E(\delta(b-Y)) = P(Y \leq b)$$

$$\text{所以 } P(X \leq a \text{ 且 } Y \leq b) = P(X \leq a)P(Y \leq b) \quad \forall a, b$$

$$\text{则 } P(x_1 < X \leq x_2 \text{ 且 } y_1 < Y \leq y_2) = P(X \leq x_2 \text{ 且 } Y \leq y_2) - P(X \leq x_1 \text{ 且 } Y \leq y_2) - P(X \leq x_2 \text{ 且 } Y \leq y_1)$$

$$+ P(X \leq x_1 \text{ 且 } Y \leq y_1) = P(X \leq x_2)P(Y \leq y_2) - P(X \leq x_1)P(Y \leq y_2) - P(X \leq x_2)P(Y \leq y_1)$$

$$+ P(X \leq x_1)P(Y \leq y_1) = (P(X \leq x_2) - P(X \leq x_1))(P(Y \leq y_2) - P(Y \leq y_1)) = P(x_1 < X \leq x_2)P(y_1 < Y \leq y_2)$$

所以 X 与 Y 独立

$$\text{若 } X \text{ 与 } Y \text{ 独立, 则 } P(X \leq a \text{ 且 } Y \leq b) = P(X \leq a)P(Y \leq b) \quad \forall a, b$$

$$\text{于是 } E(\delta(a-X)\delta(b-Y)) = E(\delta(a-X))E(\delta(b-Y))$$

$$34. \text{ 设随机变量 } X_i = \begin{cases} 1, & \text{如果车在第 } i \text{ 个站停了} \\ 0, & \text{其它} \end{cases}$$

$$\text{停站次数的期望为 } E = \sum_{i=1}^7 E(X_i)$$

$$E(X_i) = P(X_i=1) = 1 - \left(\frac{6}{7}\right)^{25} \Rightarrow E = 7 \left(1 - \left(\frac{6}{7}\right)^{25}\right)$$

35. 我们考虑每一个阴性者, 它能为“第一个阳性前的阴性人数”的期望贡献多少

单独把这个阴性者和四个阳性者拿出来. 在所有排列中, 这个阴性者排在

第一个阳性者之前的概率是 $\frac{1}{5}$. 所以每个人贡献 $\frac{1}{5}$. 最后答案是 $\frac{46}{5}$

$$36. P(\min\{X_1, \dots, X_m\} \leq k) = 1 - P(X_1 > k \text{ 且 } \dots \text{ 且 } X_m > k) = 1 - \left(\sum_{i=k+1}^{+\infty} p_i\right)^m$$

$$E(\min\{X_1, \dots, X_m\}) = \sum_{k=0}^{+\infty} k P(\min\{X_1, \dots, X_m\} = k) = \sum_{k=1}^{+\infty} \sum_{n=1}^k P(\min\{X_1, \dots, X_m\} = k)$$

$$= \sum_{n=1}^{+\infty} \sum_{k=n}^{+\infty} P(\min\{X_1, \dots, X_m\} = k) = \sum_{n=1}^{+\infty} \left(1 - P(\min\{X_1, \dots, X_m\} \leq n-1)\right) = \sum_{n=1}^{+\infty} \left(\sum_{i=n}^{+\infty} p_i\right)^m$$

37. 设 $Z_1 = \max\{X_1, \dots, X_n\}$. 则 $P(Z_1 \leq z) = P(X_1 \leq z \text{ 且 } \dots \text{ 且 } X_n \leq z) = z^n$

求导得 $p_{Z_1}(z) = nz^{n-1}$. 于是 $E(Z_1) = \int_0^1 z \cdot nz^{n-1} dz = \frac{n}{n+1}$

设 $Z_2 = \min\{X_1, \dots, X_n\}$. 则 $1 - Z_2 = \max\{1 - X_1, \dots, 1 - X_n\}$. 故 $1 - E(Z_2) = \frac{n}{n+1}$. $E(Z_2) = \frac{1}{n+1}$

极差为 $Z_2 - Z_1$. 故 $E(Z_2 - Z_1) = \frac{n-1}{n+1}$

38. 设函数 $I(x) = \begin{cases} 1, & \text{当 } x \geq \varepsilon \text{ 时} \\ 0, & \text{当 } x < \varepsilon \text{ 时} \end{cases}$. 则 $e^{a\varepsilon} I(x) \leq e^{ax}$. 两边取期望得待证式

39. $Y - X \sim N(\mu_2 - \mu_1, \sigma_1^2 + \sigma_2^2)$. 故 $P(X > Y) = P\left(\frac{Y - X - (\mu_2 - \mu_1)}{\sqrt{\sigma_1^2 + \sigma_2^2}} < \frac{\mu_1 - \mu_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}\right) = \Phi\left(\frac{\mu_1 - \mu_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}\right)$

40. $P(X=2)=0.27$ $P(Y \geq 2)=0.53$ $P(X=Y)=0.3$ $P(X \leq 2, Y \leq 2)=0.69$ $P(X > Y)=0.25$

$$41. 1 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p(x, y) dx dy = \int_0^2 dx \int_0^1 cy^2 dy = \frac{2}{3}c \Rightarrow c = \frac{3}{2}$$

$$P(X \leq 1) = \int_0^1 dx \int_0^1 dy \cdot \frac{3}{2}y^2 = \frac{1}{2}, \quad P(X+Y > 2) = \int_0^1 dy \int_{2-y}^2 dx \cdot \frac{3}{2}y^2 = \frac{3}{8}$$

$P(X=3Y)=0$ (因为它是一条线, 没有面积呀)

42. 如果人数为 n , 则营业额的期望是 $50n$

顾客花的钱和人数是独立的. 所以对上式取期望. 得 60000

其实就是 $E(\text{营业额} | \text{人数} = n) = 50n$, 然后

$$E(\text{营业额}) = \sum E(\text{营业额} | \text{人数} = n) P(\text{人数} = n) = 50 \sum n P(\text{人数} = n) = 50 E(\text{人数})$$

43. (1) 我们来老老实实算一下. 感觉计算也是一种重要的能力

设 $Z = \min\{X, Y\}$. $V = X - Y$

$$P(Z = k) = P(X = k)P(Y \geq k) + P(X \geq k)P(Y = k) - P(X = k)P(Y = k) = pq^{2k}(2-p).$$

$$l \geq 0 \text{ 时 } P(V = l) = \sum_{n=0}^{+\infty} P(Y = n)P(X = n+l) = \frac{p^2 q^l}{1-q^2}$$

$$P(Z = k \text{ 且 } V = l) = P(Y = k \text{ 且 } X = k+l) = pq^k \cdot pq^{k+l} = P(Z = k)P(V = l)$$

$$l < 0 \text{ 时 } P(V = l) = \sum_{n=0}^{+\infty} P(Y = n-l)P(X = n) = \frac{p^2 q^{-l}}{1-q^2}$$

$$P(Z=k \text{ 且 } V=l) = P(X=k \text{ 且 } Y=k-l) = pq^k \cdot pq^{k-l} = P(Z=k)P(V=l)$$

$$(2) P(W=0) = \sum_{n=0}^{+\infty} P(X=n)P(Y=n) = \frac{p^2}{1-q^2}$$

$$P(W=k) = \sum_{n=0}^{+\infty} P(X=k+n)P(Y=n) + P(X=n)P(Y=k+n) = \frac{2p^2q^n}{1-q^2}$$

$$P(Z=k \text{ 且 } W=0) = P(X=k \text{ 且 } Y=k) = p^2 q^{2k} = P(Z=k)P(W=0)$$

$$P(Z=k \text{ 且 } W=l) = P(X=k+l, Y=k) + P(X=k, Y=k+l) = 2p^2 q^{2k+l} = P(Z=k)P(W=l)$$

$$44. E(X) = \frac{1}{2}, E(Y) = \int_0^1 |x-a| dx = \int_0^a (a-x) dx + \int_a^1 (x-a) dx = \frac{2a^2-2a+1}{2}$$

$$E(XY) = \int_0^1 x|x-a| dx = \int_0^a x(a-x) dx + \int_a^1 x(x-a) dx = \frac{2a^3-3a+2}{6}$$

$$E(XY) = E(X)E(Y) \Rightarrow a = \frac{1}{2}$$

$$45. P_X(x) = \int_0^x p_{X,Y}(x,y) dy = 3x^2, P_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{P_X(x)} = \frac{1}{x}$$

$$P_Y(y) = \int_y^1 p_{X,Y}(x,y) dx = \frac{3}{2}(1-y^2), P_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{P_Y(y)} = \frac{2x}{1-y^2}$$

$$46. P_X(x) = \int_{x^2}^1 p_{X,Y}(x,y) dy = \frac{21}{8}x^2(1-x^4), P_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{P_X(x)} = \frac{2y}{1-x^4}$$

$$\text{因此 } P(Y \geq 0.75 | X=0.5) = \int_{0.75}^1 P_{Y|X}(y|0.5) dy = \frac{7}{15}$$

$$47. P_Y(y) = \int_y^1 p_{X,Y}(x,y) dx = 12y(1-y)^2, P_{X|Y}(x|y) = \frac{2(1-x)}{(1-y)^2}$$

$$E(X|Y=y) = \int_y^1 x P_{X|Y}(x|y) dx = \int_y^1 \frac{2x(1-x)}{(1-y)^2} dx = \frac{2y+1}{3}$$

$$48. E(Z|Y=y) = \int_0^y by P(X=x) dx + \int_y^{+\infty} (3x+1) P(X=x) dx = \int_0^y by \cdot \lambda e^{-\lambda x} dx + \int_y^{+\infty} (3x+1) \lambda e^{-\lambda x} dx$$

$$= by(1-e^{-\lambda y}) + \int_y^{+\infty} -(3x+1) d(e^{-\lambda x}) = by + (1 + \frac{3}{\lambda} - 3y)e^{-\lambda y}$$

$$E(Z) = \int_0^{+\infty} E(Z|Y=y) P(Y=y) dy = \int_0^{+\infty} b\lambda y e^{-\lambda y} + (\lambda + 3 - 3\lambda y) e^{-2\lambda y} dy$$

$$= \int_0^{+\infty} (-by) d(e^{-\lambda y}) + \int_0^{+\infty} \frac{\lambda+3-3\lambda y}{-2\lambda} d(e^{-2\lambda y})$$

$$= -bye^{-\lambda y} \Big|_0^{+\infty} + b \int_0^{+\infty} e^{-\lambda y} dy + \frac{\lambda+3-3\lambda y}{-2\lambda} e^{-2\lambda y} \Big|_0^{+\infty} - \int_0^{+\infty} \frac{3}{2} e^{-2\lambda y} dy$$

$$= 0 + \frac{b}{\lambda} + \frac{\lambda+3}{2\lambda} - \frac{3}{4\lambda} = \frac{2\lambda+21}{4\lambda}$$

$$49. \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} t_i t_j = \sum_{i=1}^n \sum_{j=1}^n \text{cov}(x_i, x_j) t_i t_j = \text{cov}\left(\sum_{i=1}^n t_i x_i, \sum_{i=1}^n t_i x_i\right) \geq 0$$

50. $X = e^{-\frac{U^2+V^2}{2}}$. $\sin(2Y-\pi) = \frac{V}{\sqrt{U^2+V^2}}$, $\cos(2Y-\pi) = \frac{U}{\sqrt{U^2+V^2}}$. 即 Y 也是唯一确定的

$$\frac{D(u,v)}{D(x,y)} = \begin{vmatrix} \frac{\cos(2Y-\pi)}{2\sqrt{-2\ln X}} \cdot \left(-\frac{2}{X}\right) & -2\sqrt{-2\ln X} \sin(2Y-\pi) \\ \frac{\sin(2Y-\pi)}{2\sqrt{-2\ln X}} \cdot \left(-\frac{2}{X}\right) & 2\sqrt{-2\ln X} \cos(2Y-\pi) \end{vmatrix} = -\frac{2}{X} \quad \frac{D(x,y)}{D(u,v)} = 1 / \frac{D(u,v)}{D(x,y)} = -\frac{X}{2}$$

故 $P_{U,V}(u,v) = P_{X,Y}(x,y) \left| \frac{D(x,y)}{D(u,v)} \right| = \frac{1}{2} e^{-\frac{u^2+v^2}{2}}$

51. $P(X \leq z | M) = \frac{P(X \leq z \text{ 且 } U \leq r(x))}{P(U \leq r(x))} = \frac{\int_{-\infty}^{+\infty} P(X \leq z \text{ 且 } U \leq r(x) | X=x) P_X(x) dx}{\int_{-\infty}^{+\infty} P(U \leq r(x) | X=x) P_X(x) dx}$
 $= \frac{\int_{-\infty}^z P(U \leq r(x) | X=x) P_X(x) dx}{\int_{-\infty}^{+\infty} P(U \leq r(x) | X=x) P_X(x) dx} = \frac{\int_{-\infty}^z r(x) P_X(x) dx}{\int_{-\infty}^{+\infty} r(x) P_X(x) dx} = \frac{\int_{-\infty}^z a q(x) dx}{\int_{-\infty}^{+\infty} a q(x) dx} = \int_{-\infty}^z q(x) dx$

52. $P_X(x) = \begin{cases} e^{-x}, & \text{当 } x \geq 0 \text{ 时} \\ 0, & \text{当 } x < 0 \text{ 时} \end{cases}$ 令 $Z = X+Y$. 则 $z < 0$ 时 $P_Z(z) = 0$

$z \geq 0$ 时 $P_Z(z) = \int_{-\infty}^{+\infty} P_X(u) P_Y(z-u) du = \int_0^z e^{-u} \cdot e^{-(z-u)} du = ze^{-z}$

$F(z) = \int_0^z te^{-t} dt = -ze^{-z} + 1 - e^{-z} = 1 - (z+1)e^{-z}$

$\text{med}(X) = \text{med}(Y) = -\ln \frac{1}{2}$. 但 $\text{med}(Z) \neq -2\ln \frac{1}{2}$

53. $\text{mode}(X) = \text{mode}(Y) = 0$. $\text{mode}(X+Y) = 1$